

25. The mass m of a vibrating string can be determined using the following relationship.

$$f = \sqrt{\frac{T}{4mL}}$$

where f is the fundamental frequency

T is the tension

L is the length of the string.

For a particular string the following measurements are recorded:

$$f = 110 \text{ Hz}$$

$$T = 92 \text{ N}$$

$$L = 0.63 \text{ m.}$$

Based on this information the mass of this string is

- A $3.0 \times 10^{-3} \text{ kg}$
- B $1.2 \times 10^{-2} \text{ kg}$
- C $3.3 \times 10^{-1} \text{ kg}$
- D $5.8 \times 10^{-1} \text{ kg}$
- E $3.3 \times 10^2 \text{ kg.}$